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Phantasos-Lite — Experiment 1
Results Log: Qwen2.5-7B-Instruct

Reference doc for all numbers and findings from the K=16 Qwen2.5-7B-Instruct run, kept as the baseline to compare Experiment 2 (Qwen2.5-1.5B-Instruct) and Experiment 3 (Llama-3.2-3B-Instruct, if run) against. Every number below is taken directly from notebook output, not recalculated or rounded differently.

Model & setup

Model	`Qwen/Qwen2.5-7B-Instruct`
hidden_size	3584
num_hidden_layers	28
`PRIMARY_LAYERS`	[7, 13, 21] (~25/50/75% depth)
Personas (K)	16
I-set prompts	8
Checkpoint folder	`/content/drive/MyDrive/phantasos_K16lite_checkpoints`

Persona set

8 original (pilot) + 8 added for Lite: `assistant, physician, professor, teacher, interrogator, critic, pirate, pirate_story` (original) + `customer_service, drill_sergeant, poet, poet_story, therapist, comedian, librarian, diplomat` (new).

Cluster counts: anchor=1, expert=3, warm=3, cold=4, playful=5. Pair-type counts: close=5, far=3, speaker_mode=2, surprise=4.

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1. Environment checks — all passed

CHECK	RESULT
Hook output type	`torch.Tensor` (not tuple)

Digit token ids	`[16,17,18,19,20,21,22,23,24]` — all 9 distinct, single-token
Chat template tail	correct (ends at assistant-turn open tag)
alpha=0 no-op (Check D1)	**True**
Per-position delta norm (Check D2)	last position ≈298 (expected ~298.26), all earlier ≈0

2. S2 — persona expression (judge-scored, 0–3 scale)

PERSONA	MEAN	N
assistant	1.44	16
physician	1.88	16
professor	1.94	16
teacher	1.75	16
interrogator	0.69	16
critic	1.69	16
pirate	2.56	16
pirate_story	1.81	16
customer_service	1.12	16
drill_sergeant	1.62	16
poet	2.25	16
poet_story	1.62	16
therapist	1.81	16
comedian	1.75	16
librarian	1.25	16
diplomat	1.44	16

- **S2 pass rate, excluding interrogator:** 0.69 (n=224) — threshold was ≥0.85

- **S2 pass rate, all role personas:** 0.66 — threshold was ≥0.85
- **Interpretation:** not a bug — the judge's "clearly in-role" bar under-rewards quieter, professionally-registered personas (customer_service, librarian, diplomat, interrogator) even when manual eyeballing confirms they're correctly in character. Matches and extends the same pattern found in the original K=8 pilot.
- **Length–expression correlation:** r = 0.43 (longer responses → somewhat higher judged expression score, a real but modest confound to note in Limitations)

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3. S5 — activation separability (leave-one-variant-out, layer 13)

- **Accuracy: 0.906** (chance = 0.0625 for K=16)
- Down from the K=8 pilot's 100%, as expected given 4 deliberately confusable "surprise" personas were added.
- **Confusions (all at layer 13):**
  - `drill_sergeant` → confused with `interrogator` (5/8 misclassified)
  - `librarian` → confused with `therapist` (2/8 misclassified)
  - `physician` → confused with `therapist` (5/8 misclassified)
- **Interpretation:** confusions land exactly on the engineered "surprise" boundary cases, not randomly — evidence the representation has real fine-grained structure, not just coarse categorical separation.

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4. Baseline RSA — activation vs. prompt text (embedding baseline)

Not the finding — this is the confound being controlled for.

LAYER	RSA(ACT, TEXT)	PERM-P
7	0.622	0.0002
13	0.463	0.0002
21	0.404	0.0004

- **Shuffled-label control RSA (layer 13):** −0.047 (expected ≈0 — passed)
- **Split-half noise ceiling (variant 0 vs. variant 1):**
  - layer 7: 0.769
  - layer 13: 0.837
  - layer 21: 0.967

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5. RDM-S validity (judge-free pairwise self-rating)

- 256/256 ratings collected (16×16, including diagonal self-ratings)
- **Low-confidence ratings (digit_mass < 0.8):** 0 / 256
- **Self-rating EV (expected ≈9):** mean = 8.87, min = 6.97

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6. PRIMARY RESULT — activation ↔ self-report correspondence

This is the paper's headline finding.

LAYER	SELF-DESC RSA (RAW)	SELF-DESC (PARTIAL-OUT TEXT)	SELF-RATING RSA (RAW)	SELF-RATING (PARTIAL-OUT TEXT)
7	0.464 (p=0.0002)	0.341	0.426 (p=0.0004)	0.279
13	0.616 (p=0.0002)	0.549	0.650 (p=0.0002)	0.586
21	**0.725** (p=0.0002)	**0.683**	**0.779** (p=0.0002)	**0.743**

- **RDM-F / RDM-S channel agreement:** 0.723 (two independently-constructed self-report channels — judge-scored free text and judge-free forced-digit rating — converge strongly)
- **Pattern:** layer-21-peaking, replicates the K=8 pilot's shape (0.75/0.77) at double the scale, survives partial-RSA controlling for text-wording confound, and channel agreement is *higher* than the pilot's (0.723 vs. 0.58).

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7. Bugs encountered this run (for Methods/reproducibility notes, not results)

- Activation-extraction checkpoint desync: `act_index.jsonl` claimed 256 jobs done while `act_vectors.npz` held 0 vectors (runtime interruption between index-write and the final `np.savez`). Fixed by cross-checking the index against actual stored vectors rather than trusting the index alone; re-extraction took 125s for all 256 jobs.
- `rating_readout` / `CONTRAST_LAYER` / `v_pirate` NameErrors: leftover cells from the pilot's steering section (S7/S8), which Lite correctly excludes since it's correlational-only.

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8. Files on Drive (checkpoint folder)

`s2_descriptions.jsonl`, `s2_judge.jsonl`, `act_index.jsonl`, `act_vectors.npz`, `rdm_s_ratings.jsonl` — all confirmed complete (256/256/256/768/256 respectively). Figures exported separately as PDF via direct download (not Drive) due to a storage-quota issue partway through the figure-generation pass.

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9. One-paragraph summary (for quick reference before Experiments 2/3)

Qwen2.5-7B-Instruct, K=16 personas (8 deliberately confusable), shows a strong, layer-21-peaking correspondence between internal activation geometry and self-report (RSA 0.725 self-description, 0.779 self-rating), which survives controlling for prompt-text wording and is corroborated by two independent, convergent self-report channels (agreement 0.723). Separability is high (90.6%) but blurs exactly at the engineered confusable pairs, consistent with the representation having real sub-cluster structure rather than just coarse categorical separation. S2 judge pass rate under-reports quieter personas as a known, explained limitation, not a validity problem for the primary result.